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| Course Code<br>DSN4091                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Capstone Project - Phase 1           | Course Type                                        | PJ                                         |                                |                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                      | Credits                                            | 3                                          |                                |                          |
| Prerequisite: The course to be registered after completing the third of year of undergraduate studies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                      |                                                    |                                            |                                |                          |
| Course Objectives:<br>To apply the knowledge acquired in studies in to real time problem solving in the field engineering and technology to produce the outcomes for the betterment of society.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                      |                                                    |                                            |                                |                          |
| Course Outcomes:<br>At the end of this course, the students will be able to<br>1. Explore the problems related to society/ technology<br>2. Explore the tools and technologies to develop the project<br>3. Development of foundation skills<br>4. Analyze the Suitable solutions<br>5. Preparing financial planning and interacting with subject experts with ethics and social values in individual or team.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                      |                                                    |                                            |                                |                          |
| Student Outcomes (SO): 1, 2, 3, 4, 5, 6, 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                      |                                                    |                                            |                                |                          |
| Guidelines:<br><ul style="list-style-type: none"><li>• The capstone projects can either be done at Inhouse or Industry.</li><li>• The industry project to be carried out in the industries approved by the University.</li><li>• The duration of the capstone project is 1 semester (6 Months).</li><li>• The capstone project to be carried out under supervision of a Supervisor from Industry or University.</li><li>• Students need to prepare the capstone project report as per the University format with Certificates and other necessary documents. The same to be submitted to the Office of the Concerned School after the completion of the capstone project.</li><li>• The students need to attend the reviews conducted by the University for capstone project.</li><li>• The schedule of the reviews will be announced by the School.</li></ul> |                                      |                                                    |                                            |                                |                          |
| Mode of Evaluation:<br>The four reviews will be conducted by the Panel constituted by the School as per University Norms.<br>Review 0: This is zeroth review and it will be conducted only for screening the problem statements. It does not carry any scores. It will be conducted in at the beginning of the Semester after the capstone project registration.<br><br>Review 1: This is first review will be conducted at the start of the semester to check the progress of the capstone project work. It will be conducted for 50 marks as per below rubrics.                                                                                                                                                                                                                                                                                              |                                      |                                                    |                                            |                                |                          |
| Clarity and significance of the problem statement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Feasibility and scope of the project | Quality of initial prototypes or research findings | Adherence to the project plan and timeline | Research and literature review | Methodology and approach |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5                                    | 10                                                 | 5                                          | 15                             | 10                       |

**Review 2:** This is second review will be conducted at the end of the semester to check the progress of the capstone project work. It will include Initial Project Implementation and Documentation. It will be conducted for 50 marks as per below rubrics.

| Clarity and organization of presentation materials | Communication and collaboration within the team | Quality of the Phase - 1 written project report | Demonstration of key project components | Ability to address questions and provide explanations |
|----------------------------------------------------|-------------------------------------------------|-------------------------------------------------|-----------------------------------------|-------------------------------------------------------|
| 10                                                 | 10                                              | 15                                              | 10                                      | 05                                                    |

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| <i>Recommendation by the Board of Studies on</i> | 16.04.2024     |
| <i>Approval by Academic council on</i>           | 23.05.2024     |
| <i>Compiled by</i>                               | Dr. Sandip Mal |